

Homework 4

Solve the following problems justifying all your answers. You should not use a graph calculator to justify any answer.

1. Consider the function $g(x) = x + |x|$.
 - (a) For what value of x is this function differentiable?
 - (b) Find a formula for g' using the definition of a derivative.
2. Find all points (x, y) on the graph of $f(x) = 3x^2 - 4x$ with tangent lines parallel to the line $y = 8x + 5$.
3. Let $f(x) = (1 + x^2)(x^{\frac{3}{4}} - x^{-3})$. Find $f'(x)$ in two different ways:
 - (a) By applying the product rule.
 - (b) By multiplying the factors to produce a sum of simpler terms to differentiate.
4. Consider the function $f(x) = x^2 e^x$.
 - (a) Find expressions for the first five derivatives of $f(x)$.
 - (b) Guess a formula of $f^{(n)}(x)$ and prove it using mathematical induction for all $n \in \mathbb{N}$.
5. Prove that a function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f'(x) = cf(x)$ for some $c \in \mathbb{R}, c \neq 0$, if and only if $f(x) = ke^{cx}$ for some $k \in \mathbb{R}$.